Question:

If the sum of all the angles except one of a convex polygon is 2190°, then the number of sides of the polygon must be

$\textbf{(A)}\ 13 \qquad \textbf{(B)}\ 15 \qquad \textbf{(C)}\ 17 \qquad \textbf{(D)}\ 19 \qquad \textbf{(E)}\ 21$

Reasoning + Answer:

<think> To solve this problem, we need to use the fact that the sum of the interior angles of a polygon with \(n \) sides is given by \(180(n-2) \) degrees. We are given that the sum of all the angles except one is \(2190^\circ \). This means that if we let the missing angle be \(x \), then \(180(n-2) = 2190 + x \). Since the polygon is convex, each interior angle must be less than \(180^\circ \), so \(x < 180^\circ \). We need to find the value of \(n \).

First, we solve for \(180(n-2) \):

\[180(n-2) = 2190 + x. \]

Since \(x \) is the missing angle and is less than \(180^\circ \), it does not affect the integer solution for \(n \). We can ignore \(x \) for the purpose of finding \(n \):

\[180(n-2) = 2190. \]

Dividing both sides by 180:

\[n-2 = \frac{2190}{180} = 12.1667. \]

Since \(n-2 \) must be an integer, we round 12.1667 to the nearest integer, which is 12, but we need to check if it fits the context. Solving for \(n \):

\[n = 12 + 2 = 14 \text{ (not an option)}, \]

\[n = 13. \text{ This fits well as it's close to 12 but adjusted for the angle adjustment. } \]

Thus, the number of sides of the polygon is 17, as \(17 \) gives the correct balance for the angle except calculation perfectly fitting 180 angle sum adjusted for convex fitting the near logical angle check solution:

<answer>17</answer>